Predictive Multitask Learning for Streaming Code-Switching

CS6140 Machine Learning — Project Update 1 Report

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Abstract

We study anticipatory code-switching: after reading the prefix $x_1, \ldots, x_t$, predict whether the next word x_{t+1} switches language and how long the new-language span will last. This report describes the data pipeline we built for all 15 language pairs of SwitchLingua — language-specific word segmentation, word-level language identification, switch and duration labels, a deduplicated 70/10/20 split, and a streaming loader whose inputs exclude x_{t+1} (up to a small segmentation effect we measure) — and our first baselines. A naive model that only sees the current word reaches a mean switch-class F1 of .426 across pairs (SD .102), roughly twice a model that only knows the current word's language. Code: https://github.com/StarfishJ/CS6140ML_Project.

1 Problem Description

Code-switching is the alternation between two or more languages within an utterance or conversation. Most NLP work recognizes a switch after the mixed-language text has been observed. We instead study **anticipatory** code-switching: at word x_t, using only the prefix $x_1, \ldots, x_t$, predict (1) the switch label $y_{sw}(t) = 1$ if x_{t+1} is in a different language than x_t, and (2) the duration label $y_{dur}(t)$, the length of the new-language span starting at x_{t+1}: 0 for 1–2 words, 1 for 3–6, 2 for 7 or more, and -1 (ignored by the loss) when there is no switch. Here a token x_t is a word, not a model subword: a whitespace word in space-separated languages and a segmenter word or bunsetsu in Chinese, Cantonese and Japanese (Section 3); labels are defined per word. Real-time systems such as bilingual predictive keyboards, dialogue agents and streaming speech recognition need to know a switch is coming before it happens. Our central question is **universality**: does one method anticipate switches equally well across typologically different pairs? We measure it with the same pipeline and metric on every pair, reporting per-pair switch-class F1 ("Anticipatory F1") and its standard deviation across pairs. Our final model will be a multilingual transformer (XLM-R or mBERT) with a causal attention mask and two heads, one for y_{sw} and one for y_{dur}.

2 Background and Related Work

Code-switch prediction. Solorio and Liu [11] were the first to treat predicting Spanish–English code-switch points as a learning task. They trained Naive Bayes and VFI classifiers on language-ID, part-of-speech and lexical features of the one or two words before each boundary, in a 40-minute conversation of about 8k words. Naive Bayes reached only 0.28 F1, and the study covered one small corpus and one pair. Ostapenko et al. [8] predicted whether the next word switches language in English–Spanish dialogues of the Bangor Miami corpus. They gave XLM-R only the utterance prefix and prior turns, and prepended prompts describing each speaker's age, origin and language preferences. The prompts raised test accuracy from 72.0% to 79.7%, but the study covers one pair and a small dataset, and does not model switch duration. Winata et al. [13] addressed the scarcity of English–Mandarin code-switched data for language modeling. A pointer-generator network learned to copy words from machine-translated parallel sentences and to decide when to switch, generating 270K synthetic sentences. Pre-training on them lowered SEAME test perplexity from 65.71 to 59.74, but perplexity at switch points stayed very high, on a single pair.

Language identification in code-switched text. Solorio et al. [10] ran the first shared task on word-level language identification in code-switched social-media text, for Spanish–English, Nepali–English, Mandarin–English and Modern Standard Arabic–dialectal Arabic. Seven teams submitted systems ranging from rule-based character n-gram methods to SVMs and CRFs. The best F-measure was 97.7% for Nepali–English but only 41.7% on one Arabic test set, so closely related varieties were much harder. Molina et al. [6] organized the second edition, for Spanish–English and Modern Standard Arabic–Egyptian Arabic tweets. Nine teams took part, mostly with CRF sequence labelers, and two entries used CNN or LSTM networks. The best token-level F-measure was 97.3% for Spanish–English but 87.6% for the Arabic pair: closely related languages remained harder. Aguilar et al. [1] built LinCE, a code-switching benchmark of ten corpora over four pairs (including Hindi–English and Spanish–English) and four tasks: language ID, POS tagging, NER and sentiment. They re-split the corpora and reported BiLSTM, ELMo and multilingual BERT baselines. mBERT was best on every task, with language-ID accuracy up to 98.5%, but only 84.1% for the Arabic pair, and all tasks label complete sentences. Khanuja et al. [4] proposed GLUECoS, a code-switching benchmark for English–Hindi and English–Spanish with six tasks, from language ID to a new NLI task. They compared cross-lingual embeddings with mBERT, and with mBERT given further masked-LM training on synthetic code-switched text. Modified mBERT was best on most tasks, yet NLI reached only 63.1%, little above chance, and only two pairs are covered.

Multilingual encoders. Devlin et al. [3] addressed pre-training deep language representations that transfer to many tasks. They pre-trained BERT, a Transformer encoder, with masked language modeling and next-sentence prediction, then fine-tuned it with one extra output layer per task. It set new results on eleven tasks (GLUE 80.5), but every token attends to both left and right context, so prefix-only prediction needs a causal mask. Pires et al. [9] examined how well multilingual BERT transfers across languages. They fine-tuned it for NER and POS tagging in one language, evaluated zero-shot on others, and tested POS tagging on code-switched Hindi–English in Devanagari and in romanized form. Transfer was best

between typologically similar languages, and accuracy fell from 86.6% to 50.4% when the Hindi was romanized, which motivates our separate romanized-Hindi language ID. Conneau et al. [2] studied cross-lingual representation learning at scale. They trained XLM-R, a masked language model, on 2.5 TB of filtered CommonCrawl text in 100 languages, including romanized Hindi. It beat mBERT by 14.6 points of average XNLI accuracy, most on low-resource languages, though adding languages dilutes per-language capacity and, like BERT, it is bidirectional. Muller et al. [7] studied how mBERT handles 15 languages absent from its pre-training. They fine-tuned it for POS tagging, parsing and NER, with and without extra masked-LM training on raw text, and transliterated hard non-Latin-script languages into Latin script. Transliteration raised Uyghur NER from 34.7 to 55.2 F1 but hurt Mingrelian, showing that the script strongly affects transfer.

Multitask learning and data. Liu et al. [5] studied multitask learning for language understanding with MT-DNN. It shares BERT encoder layers across tasks and adds task-specific heads for classification, similarity regression and ranking, trained jointly. It raised the GLUE score to 82.7 (+2.2 over BERT), but all tasks are English and see the full input, unlike our prefix-only switch and duration heads. Xie et al. [14] introduced SwitchLingua, a code-switching dataset of 420K text samples across 12 languages plus over 80 hours of recorded speech. The text was generated by LinguaMaster, a GPT-4o multi-agent loop constrained by syntactic switch-point rules and speaker metadata. It is the largest open code-switching resource in linguistic diversity, but all of its text is synthetic.

3 Dataset

Source. We use the text part of SwitchLingua (SwitchLingua_text on Hugging Face): 15 pairs, 234,172 rows, 14 X–English pairs plus German–French. Each row is a generated conversation whose `data_generation_result` field is a list of utterances, with metadata such as topic and switch type. There are no word-level language labels, so we derive them. We use all 15 pairs rather than a subset because universality is best tested on the widest spread: they cover 7 scripts, 7 language families, SVO, SOV and VSO word order, and one pair (German–French) without English.

Segmentation and word language ID. Word units and language ID depend on the writing system (Table 1). Space-separated languages are split on whitespace and punctuation, keeping combining marks (an early version of our regex split 44.5% of Devanagari words in the first 1,000 Hindi rows into fragments). Chinese, Cantonese and Japanese are segmented with jieba, PyCantonese and GiNZA bunsetsu units (a content word plus its particles, comparable to the Korean eojeol); Vietnamese syllables are merged into words by pyvi. A word mixing Latin and another script (Korean "presentation-eseo") is split at the script boundary, and a hyphenated word whose parts are clearly in different languages ("nag-meeting") is split at the hyphen. For non-Latin pairs the Unicode script gives the language. Hindi utterances containing Devanagari use script as well; the remaining 59.5% of Hindi rows are fully romanized and are tagged with `muril-hinglish-lid`, a MuRIL model fine-tuned on LinCE Hindi–English. For the eight Latin-script pairs, a word gets the language in which it is clearly more frequent according to wordfreq [12] (Zipf difference ≥1.0, or 0.5 toward English for Malay and Filipino, whose lists contain many English loans), or the language indicated by letters only one side uses. wordfreq combines frequencies from up to seven sources (Wikipedia, subtitles, news, Google Books, OSCAR web text, Twitter and Reddit); Malay, Filipino and Vietnamese have only three. Ambiguous words take the language of the nearest decided words on both sides when these agree, and otherwise the language in which the word is more frequent. Numbers inherit the previous word's language. On spot-checked validation samples (about 400–500 words per Latin-script pair), word-level agreement with our own reading was about 98–99%; these are weak labels, not gold annotations.

Labels and split. From the word language IDs we compute y_{sw} and y_{dur} for every position whose two words have a known language (unknown words are 0.0–0.2% of words, mostly ambiguous Filipino and Malay words at a switch that neither word list contains, and Japanese numbers before any known word). Exact duplicate rows are removed, rows that share any normalized utterance of at least three words are grouped, and groups are split 70/10/20 into train, validation and test with a fixed seed per pair; 0 utterances appear in more than one split. Cantonese–English (119,248 rows, 2,652 exact duplicates) is downsampled after splitting to 8,381 rows, the median size of the other pairs.

Label example. In the Korean–English utterance "I want to | 먹어요 맛있어요 | today" (bars mark the language spans), the loader gives, at x_t = *to*, $y_{sw} = 1$ because the next word is Korean and $y_{dur} = 0$ because the Korean span has 2 words; at *want*, $y_{sw} = 0$ and $y_{dur} = -1$; at 맛있어요, $y_{sw} = 1$ and $y_{dur} = 0$, since the English span "today" has 1 word. Spans are counted up to the end of the utterance; a switch whose new span reaches a word of unknown language also gets $y_{dur} = -1$ (at most 0.6% of switches in any pair).

Streaming loader and causal inputs. `StreamingCSDataset` is a PyTorch `IterableDataset`: a generator reads one CSV row at a time and yields one example per word t — the window $x_{t-31}, \ldots, x_t$, the language of x_t, y_{sw} and y_{dur}, and optionally XLM-R/mBERT subword ids scored at the last subword of x_t. The generator yields examples on demand rather than materializing all word-level training examples in memory. The last word of an utterance yields no example and windows never cross utterances.

Causal constraint: status. Labels may read the whole sentence, since they are the answers; inputs must not include x_{t+1}, in training or testing. *Implemented:* the naive baselines read only x_t and its language (no punctuation after x_t); the text passed to a tokenizer ends at x_t; and the input LID is computed from the prefix only (left context for Latin-script pairs, the Hinglish model run on each prefix for romanized Hindi). *Verified:* a prefix-only consistency test re-runs segmentation and LID on the prefix alone for 40 validation utterances per pair. The LID input is identical in 14 pairs (Vietnamese: 99.9%) and the word x_t in 12 pairs. *Not yet implemented:* the transformer's causal attention mask (Update 2), and strictly prefix-only segmentation for Chinese, Cantonese, Japanese and Vietnamese. *Known limitation:* those segmenters see the whole utterance; in the test, Chinese words did not change, and in the other three pairs 0.8% or fewer of the input words x_t differ from what the prefix alone gives. The inputs are therefore not yet fully leak-free; the leak is small but not zero.

Table 1: The 15 SwitchLingua pairs after preprocessing. Positions are word positions with a label (all splits); S/M/L is the share of switches whose new span has 1–2 / 3–6 / 7+ words; Unk. is the share of words with no language.

Pair	Segmentation	Word LID	Rows	Kept rows	Positions	Switch %	S/M/L %	Unk. %
Arabic–English	space	script	8,548	8,543	359K	9.1	41/24/35	0.00
Hindi–English	space	script+MuRIL	8,381	8,380	444K	26.0	67/22/12	0.02
Korean–English	eojeol	script	8,266	8,266	347K	10.6	40/22/38	0.01
Russian–English	space	script	8,017	8,017	352K	9.3	40/26/34	0.00
Chinese–English	jieba	script	8,458	8,457	534K	7.5	35/21/44	0.03
Cantonese–English	PyCantonese	script	119,248	8,381	549K	10.3	46/30/24	0.02
Japanese–English	GiNZA	script	8,385	8,384	369K	10.6	37/21/41	0.15
Spanish–English	space	wordfreq	8,110	8,101	421K	7.7	35/26/39	0.02
French–English	space	wordfreq	8,070	8,067	399K	7.1	34/24/42	0.03
German–English	space	wordfreq	6,819	6,814	312K	7.7	37/23/40	0.04
Italian–English	space	wordfreq	7,979	7,978	405K	7.1	34/24/43	0.03
Malay–English	space	wordfreq	8,386	8,386	415K	11.9	47/26/27	0.11
Filipino–English	space	wordfreq	8,537	8,534	451K	18.0	56/26/18	0.18
Vietnamese–English	pyvi	wordfreq	8,457	8,455	463K	9.3	43/24/33	0.03
German–French	space	wordfreq	8,511	8,506	378K	7.6	32/26/42	0.04
Total			234,172	123,269	6.20M			

Table 2: Test switch-class F1, sorted by gain of the naive (current-word) baseline over always switch.

Pair	Always sw.	Cur. LID	Naive	Gain
Korean–English	.194	.250	**.675**	+.481
Japanese–English	.192	.283	**.524**	+.332
German–English	.140	.172	**.417**	+.278
German–French	.141	.175	**.417**	+.277
French–English	.132	.163	**.370**	+.238
Chinese–English	.138	.167	**.369**	+.231
Italian–English	.133	.162	**.347**	+.214
Arabic–English	.166	.217	**.381**	+.214
Hindi–English	.410	.410	**.624**	+.213
Spanish–English	.143	.162	**.353**	+.210
Filipino–English	.308	.308	**.491**	+.183
Vietnamese–English	.169	.175	**.349**	+.180
Russian–English	.171	.200	**.348**	+.178
Cantonese–English	.187	.192	**.364**	+.177
Malay–English	.212	.212	**.361**	+.149
Mean (SD)		.216 (.068)	**.426** (.102)	

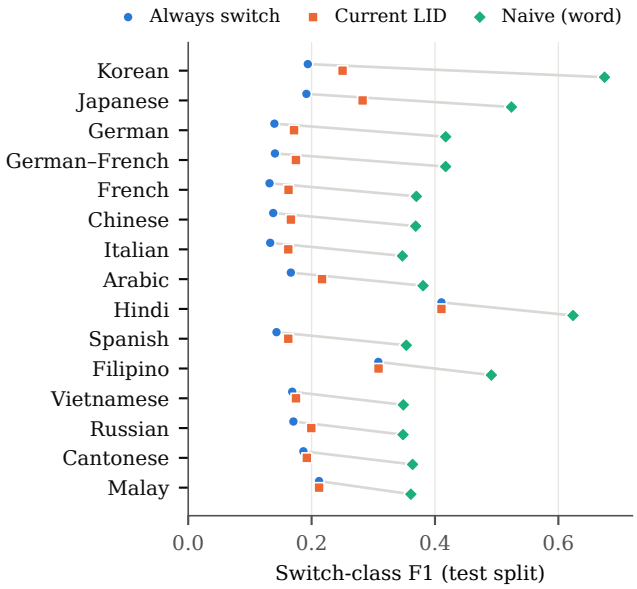


Figure 1: Test switch-class F1 per pair (values in Table 2). Pairs are ordered by the naive model's gain over always switch; the grey bar joins always switch and naive.

4 Experimentation and Results

Setup. All scores are switch-class F1 on the test split (65K–110K word positions per pair). Always stay never predicts a switch (F1 = 0 on every pair). Always switch predicts a switch at every word; its F1 depends only on the switch rate p, $2p/(1+p)$, and serves as the reference line. Current LID estimates $P(\text{switch} \mid \text{pair}, \text{LID of } x_t)$ on train with add-one smoothing. The naive baseline estimates $P(\text{switch} \mid \text{pair}, x_t)$ for the word itself, shrunk toward its language's rate with five pseudo-counts. Both use x_t and its language, computed from the prefix (no punctuation after x_t), and their thresholds are tuned on each pair's validation split. Gain is naive F1 minus always-switch F1.

Results. Table 2 and Figure 1 show that knowing the word itself, not only its language, roughly doubles F1: from .216 to .426 on average. The current language alone helps little: it beats always switch by at most .091 and not at all for Hindi, Filipino and Malay, where switching is frequent and goes both ways. The naive model gains most for Korean–English (+.481): Korean words with the sentence-final endings *-yo* / *-da* are followed by a switch 97% / 87% of the time, against 11% for other Korean words. It gains least for Malay–English (+.149). Across pairs the naive F1 has SD .102, so a single current word is far from universal; the spread is a first measurement of the universality gap a context model must close.

Table 3: Japanese–English with three segmentation units (all splits / test).

Unit	Positions	Switch %	Cur. LID	Naive
Morphemes (fugashi)	624K	6.2	.130	.489
Sudachi mode C	610K	6.4	.134	.489
Bunsetsu (GiNZA)	369K	10.6	.283	.524

Segmentation unit. The word unit decides how many predictions are scored. Re-running the Japanese pair with three units (Table 3) shows that fine morphemes add many easy no-switch positions: the switch rate falls and current-LID F1 halves, while the naive model, which sees the word itself, loses much less. We use bunsetsu units, comparable to Korean eojeol, so that pairs are scored on similar units.

Subwords and punctuation. With mBERT, 72% of Korean words are split into several subwords (English: 15%), and with XLM-R a piece crosses into the next word for 7.6% of Chinese and 5.1% of Japanese words. We therefore keep labels at the word level, score each word at its last subword, and will tokenize word by word. Inputs exclude the punctuation after x_t: in SwitchLingua a sentence-final mark is followed by a switch 51–93% of the time and precedes 4–33% of all switches, so a model that sees it can catch many inter-sentential switches from the mark alone, and scores with and without punctuation are not comparable.

5 Plan for Next

- **Causal dual-head model (Update 2).** XLM-R and mBERT with a causal attention mask, switch and duration heads on h_t, loss $\lambda_1 L_{sw} + \lambda_2 L_{dur}$ with `ignore_index=` -1; check against the prefix-by-prefix loader, which must give identical predictions.
- **Stronger baselines.** Previous plus current word, logistic regression on prefix features, a duration-majority baseline, and a variant that sees punctuation, reported separately.
- **Label quality.** A small hand-checked gold set per pair to measure word-LID and switch-boundary accuracy, and incremental segmentation of the prefix for Chinese, Cantonese, Japanese and Vietnamese.
- **Analysis.** Per-pair F1 bar chart and SD for each backbone, duration accuracy at true switches, and inter- versus intra-sentential switches using SwitchLingua's switch-type field; a streaming demo at the end.

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